

ShadowTrace AI

Behavioral Cybercrime Intelligence for Coordinated Identity Detection · CIDECODE 2026 · 3rd Edition CCITR Tech Hackathon

5	200+	87%	3	XAI
SIGNAL LAYERS	ANNOTATED POSTS	VALIDATION ACCURACY	RISK TIERS	EXPLAINABLE

OVERVIEW

Cybercrime actors frequently operate multiple aliases across social media platforms to evade detection and impersonate legitimate services. Manually correlating these accounts is slow, inconsistent, and difficult to sustain across active investigations. Unlike conventional keyword or metadata-matching tools, **ShadowTrace AI operates on behavioral patterns** — making it resilient to alias changes and account recycling. The system is a SOCMINT prototype that combines behavioral fingerprinting and Explainable AI to identify coordinated digital identities from publicly available data.

TECHNICAL APPROACH

The system extracts **five behavioral signals** across submitted handles — posting timing patterns, multilingual phrase similarity (Hindi + English fraud templates), hashtag cluster overlap, stylometric writing analysis, and repeated payment link patterns. Similarity scoring uses multilingual sentence-transformer embeddings. A weighted explainable scoring engine produces a **0–100% Coordinated Activity Score**, classified into three actionable tiers: **HIGH** (escalate), **MEDIUM** (monitor), and **LOW** (archive).

#	SIGNAL	METHOD	WEIGHT
1	Posting Timing Patterns	Cross-account temporal delta analysis	20%
2	Multilingual Phrase Similarity	Sentence-transformer embeddings (HI+EN)	25%
3	Hashtag Cluster Overlap	Jaccard similarity on tag sets	20%
4	Stylometric Analysis	Punctuation, emoji, sentence structure fingerprint	20%
5	Payment Link Patterns	Shortened URL hash matching across handles	15%

INVESTIGATOR OUTPUT

Each analysis generates a **plain-language XAI reasoning summary** explaining which signals triggered the verdict. A timestamped incident report is produced with legal references (IT Act §66C, §66D; IPC §420), relationship graph visualisation, and a recommended action — formatted for direct handoff to a cybercrime unit.

TIER	SCORE RANGE	ACTION	INDICATOR
HIGH	70 – 100%	Escalate to cybercrime unit immediately	●
MEDIUM	40 – 69%	Place under active monitoring	●
LOW	0 – 39%	Archive; flag for periodic review	●

SYSTEM INTERFACE — LIVE PROTOTYPE

The interface accepts two social media handles, runs the five-signal behavioral pipeline in real time, and renders a live **Coordinated Activity Score** with full XAI reasoning breakdown and interactive relationship graph. A downloadable timestamped incident report (.txt) is generated on completion. The demo mode runs on a pre-loaded anonymised OSINT dataset of 200 ethically collected public posts — no live API calls are required during demonstration.

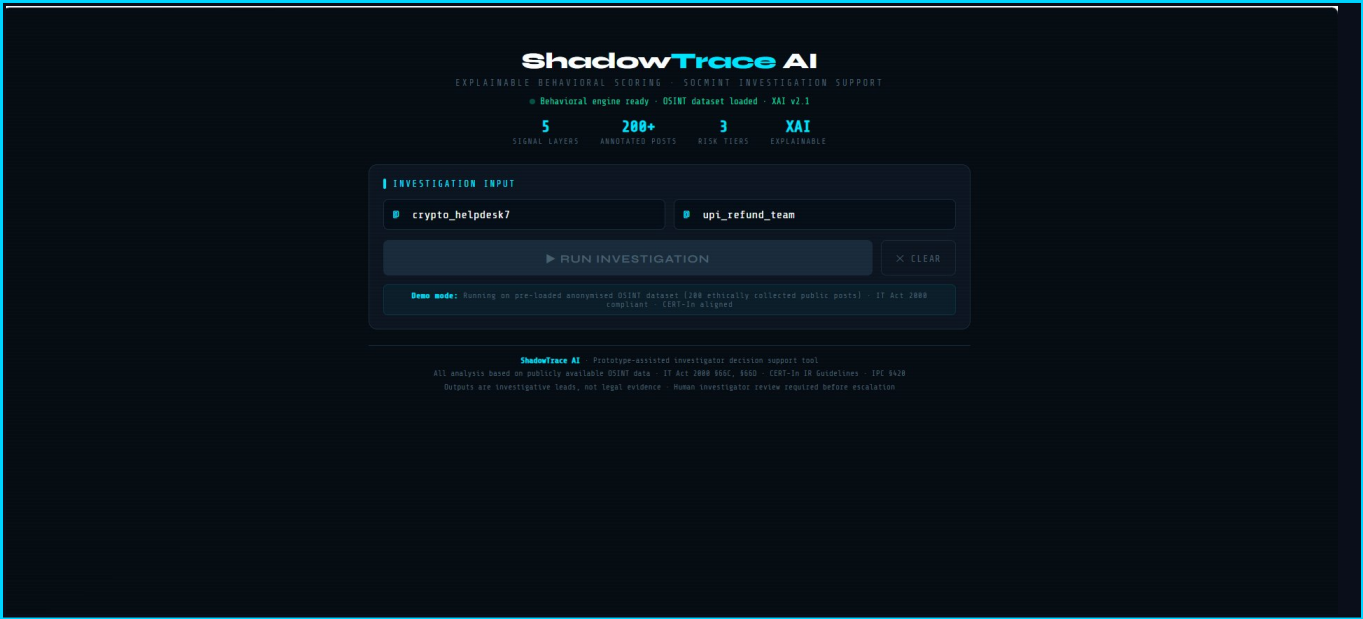


Fig 1. ShadowTrace AI live investigator interface — Investigation Input panel with behavioral engine status indicators, demo dataset confirmation, and legal compliance footer.

EXAMPLE BEHAVIORAL INDICATORS

SIGNAL	EXAMPLE INDICATOR	FRAUD VECTOR
Phrase Match	Repeated phrases: 'refund pending', 'verify now'	Phishing / Impersonation
Timing Pattern	Accounts posting within identical 5-minute windows	Coordinated Botnet
Stylometry	Consistent punctuation, emoji usage, sentence structure	Same-author fingerprint
Payment Links	Shared shortened payment URLs detected across handles	UPI / Crypto fraud
Hashtag Overlap	Identical hashtag clusters across separate accounts	Amplification network

COMPLIANCE & VALIDATION

87%	Validation Accuracy System verdicts vs. ground-truth annotations across 200 anonymised, ethically collected public posts across manually labelled account pairs. Behavioral signal extraction and similarity scoring validated through structured test protocol.
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All analysis operates exclusively on publicly available OSINT data, aligned with CERT-In Incident Response Guidelines and IT Act 2000. The scoring pipeline is deterministic, ensuring reproducibility across repeated analyses. A functional end-to-end prototype is complete and available for live demonstration.

IT Act 2000 §66C, §66D	IPC §420	CERT-In Aligned	OSINT Only	XAI Explainable
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DEPLOYMENT & DEMONSTRATION

The prototype runs entirely client-side in a browser with no backend dependency, enabling immediate deployment without infrastructure setup. The system is designed as a deployable investigator-support workflow with interactive analysis, relationship graph visualisation, deterministic behavioral scoring, and downloadable incident report generation. Analysis of a submitted account pair completes in under 3 seconds. The system is intended for demonstration in cybercrime investigation

and SOCMINT evaluation environments.

■ Live Prototype: haresh03072008-cyber.github.io/CIDECODE-Hackathon---2026/